
Create an AWS Application Load Balancer (ALB)

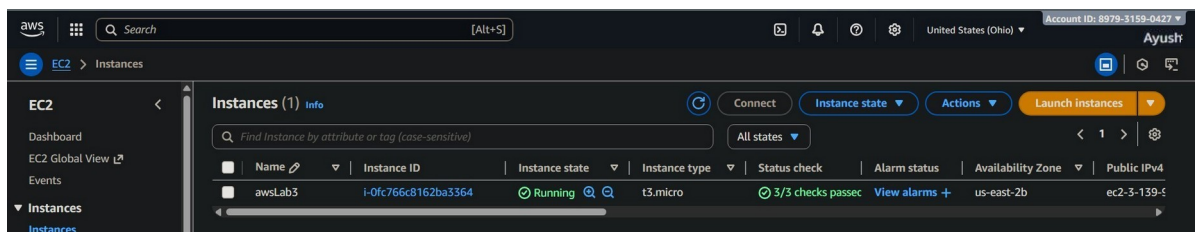
Create an Application Load Balancer that distributes incoming HTTP traffic across multiple EC2 instances.

Prerequisites

- AWS account with necessary permissions (EC2, ELB access)
 - At least two running EC2 instances in the same VPC with a security group allowing HTTP (port 80)
 - Basic understanding of AWS Console navigation
-

Step 1: Login and prepare environment

1. Login to your AWS Management Console at <https://aws.amazon.com/console/>
2. Navigate to **EC2 Dashboard** from the Services menu.



Step 2: Create Two new EC2 instances for ALB(Application Load Balancer) us

- Click on **LAUNCH INSTANCES**.
- **Fill the required data.**

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

lab5-a

Add additional tags

- Choose UBUNTU as AMI

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a search field or choose **Browse more AMIs**.

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-02d26659fd82cf299 (64-bit (x86)) / ami-0b9093ea00a0fed92 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

- Select the Instance Type (**free tier Eligible**).

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.017 USD per Hour On-Demand RHEL base pricing: 0.0268 USD per Hour

On-Demand Linux base pricing: 0.0124 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour On-Demand SUSE base pricing: 0.0124 USD per Hour

Additional costs apply for AMIs with pre-installed software

- In Network Settings Allow HTTP traffic from the internet.

▼ Network settings [Info](#)

Network [Info](#)

vpc-01f007e5fdc3847aa

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-3' with the following rules:

☒ Allow SSH traffic from
Helps you connect to your instance

Anywhere
0.0.0.0/0

☐ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

- Edit the Network Settings.

▼ Network settings [Info](#)

VPC - required | [Info](#)

vpc-01f007e5fdc3847aa (default) ▼
172.31.0.0/16

Subnet | [Info](#)

subnet-04eb704976977f7fc ▼
VPC: vpc-01f007e5fdc3847aa Owner: 919313850164
Availability Zone: ap-south-1b (aps1-az3) Zone type: Availability Zone
IP addresses available: 4091 CIDR: 172.31.0.0/20

Auto-assign public IP | [Info](#)

Enable ▼

Additional charges apply when outside of free tier allowance

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to

☒ Create security group ☐ Select existing security group

- **Visit Advanced Details.**
- Go to **User Data- optional** at the end of page.
- Paste the data in **WORKSPACE** or create a bin file with same data and Upload it.

o #!/bin/bash

Update package lists

o apt-get update

Install nginx without asking for confirmation

o apt-get install nginx -y

Set homepage to show "Hey it's <hostname>"

o echo "Hey it's \$(hostname)" > /var/www/html/index.html

- **Launch** the instance.

▼ Summary

Number of instances | [Info](#)

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-02d26659fd82cf299

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB



Free tier: In your first year of opening an AWS



[Cancel](#)

[Launch instance](#)



[Preview code](#)

● Create 2 similar instances.

Instances (2) [Info](#)

Last updated
less than a minute ago

[Connect](#)

[Instance state](#) ▼

[Actions](#) ▼

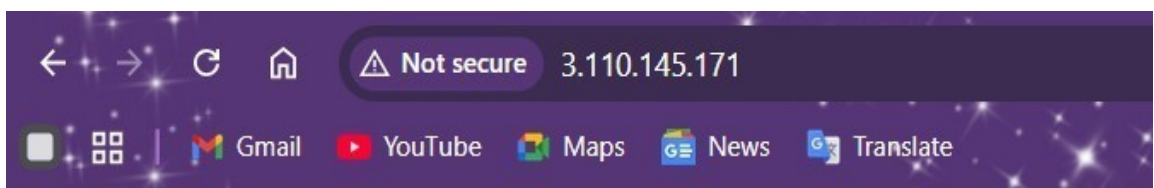
[Launch instances](#) ▼

[All states](#) ▼

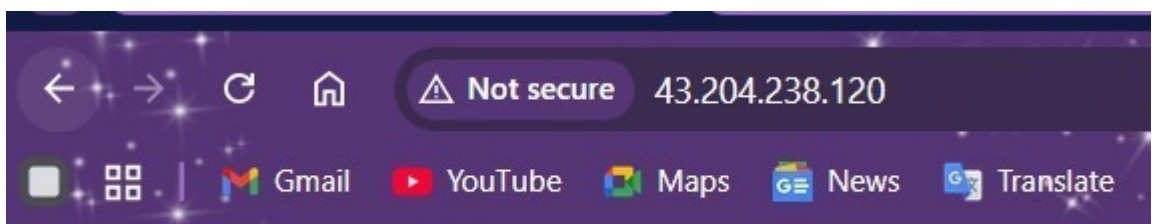
< 1 >

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☐	lab5-a	i-066f4a8b180ba3f9b	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1c	ec2-3-110-1
☐	lab5-b	i-07aa1599aaed11b2e	Running	t2.micro	Initializing	View alarms +	ap-south-1b	ec2-43-204

● Run the public Ips of two instance in new tabs.



Hey it's ip-172-31-18-94



Hey it's ip-172-31-2-173

Step 3: Create a Target Group

1. In the EC2 dashboard sidebar, locate **Load Balancing** and click **Target Groups**.

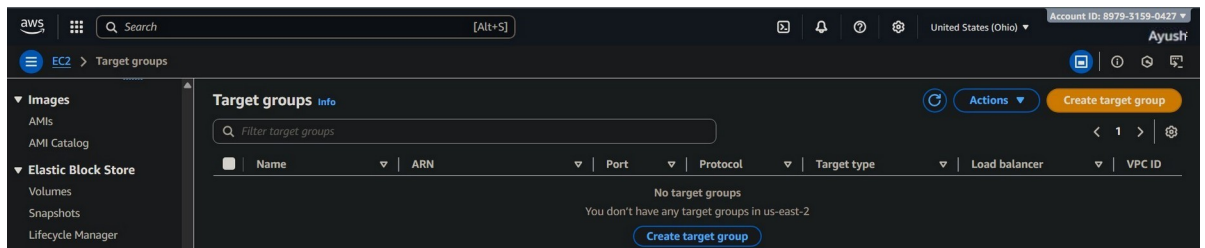
▼ Load Balancing

Load Balancers

Target Groups

Trust Stores

2. Click **Create target group**.



3. Select **Target type** as **Instance**.

Specify group details

Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Basic configuration

Settings in this section can't be changed after the target group is created.

Choose a target type

☒ Instances

- Supports load balancing to instances within a specific VPC.
- Facilitates the use of [Amazon EC2 Auto Scaling](#) to manage and scale your EC2 capacity.

4. Enter a **Name** for your target group, e.g., MyTargetGroup.
5. Set **Protocol** to HTTP and **Port** to 80.
6. Select the correct **VPC** where your EC2 instances are located.

Target group name

talb1

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol

Protocol for load balancer-to-target communication. Can't be modified after creation.

HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-01f007e5fdc3847aa
172.31.0.0/16

(default)



7. Click **Next**.

► Tags - optional

Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

Cancel

Next

8. Under **Registered targets**, select your two EC2 instances.

Register targets

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Filter instances

< 1 > ⚙

☒	Instance ID	Name	State	Security groups	Zone
☒	i-07aa1599aaed11b2e	lab5-b	Running	launch-wizard-4	ap-south-1b
☒	i-066f4a8b180ba3f9b	lab5-a	Running	launch-wizard-3	ap-south-1c

2 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

9. Click **Add to registered** > then click **Create target group**.

Review targets

Targets (2) Remove all pending

Show only pending < 1 > ⚙️

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID
i-07aa1599aaed11b2e	lab5-b	80	Running	launch-wizard-4	ap-south-1b	172.31.2.173	subnet-04eb704976977f7fc
i-066f4a8b180ba3f9b	lab5-a	80	Running	launch-wizard-3	ap-south-1c	172.31.18.94	subnet-01bdf90b1522f801b

2 pending Cancel Previous Create target group

Step 4: Create a SECURITY GROUP

1. In the **EC2** Dashboard Sidebar, Visit **Network & Security**.

▼ Network & Security

Security Groups

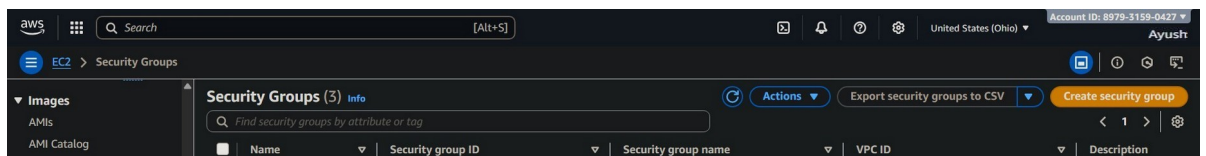
Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

2. Click on **Create Security Group**.



3. Add Security group name.
4. Fill the description-optional.
5. Select the correct **VPC**.

Basic details

Security group name [Info](#)

SecurityGroup1

Name cannot be edited after creation.

Description [Info](#)

Security for Load balancer

VPC [Info](#)

vpc-01f007e5fdc3847aa

6. ADD INBOUND RULES for SSH & HTTP with source as **Anywhere-IPv4**.

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
SSH	TCP	22	Anywh... 0.0.0.0/0	SSH Delete
HTTP	TCP	80	Anywh... 0.0.0.0/0	HTTP! Delete

[Add rule](#)

7. Click on **Create security group**.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags

[Cancel](#) [Create security group](#)

8. Verify the Security Group.

✔ Security group (sg-0b56e355a47d036a4 | SecurityGroup1) was created successfully

▶ Details

Find security groups by attribute or tag				
	Name	Security group ID	Security group name	VPC ID
☐	-	sg-09bc2950605d5f1df	default	vpc-01f007e5fdc3847aa
☒	-	sg-0b56e355a47d036a4	SecurityGroup1	vpc-01f007e5fdc3847aa

Step 5: Create the Application Load Balancer

1. In the EC2 dashboard sidebar, click **Load Balancers**.

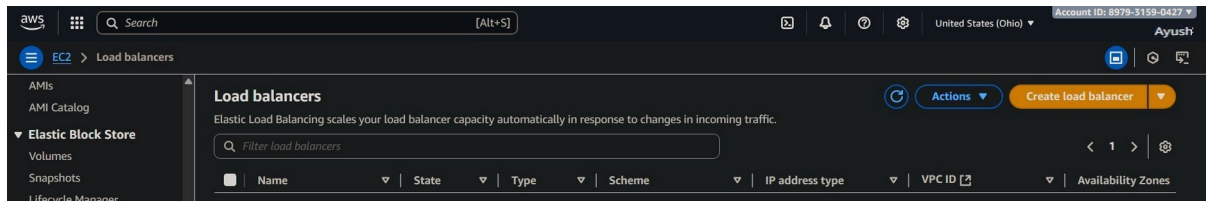
▼ Load Balancing

Load Balancers

Target Groups

Trust Stores

2. Click **Create Load Balancer**.



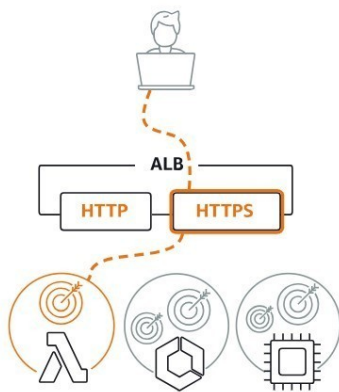
3. Select **Application Load Balancer** and click **Create**.

Compare and select load balancer type

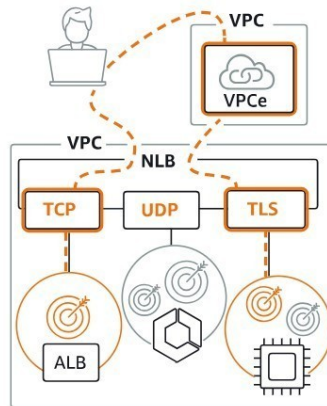
A complete feature-by-feature comparison along with detailed highlights is also available. [Learn more](#)

Load balancer types

Application Load Balancer [Info](#)



Network Load Balancer [Info](#)



Gateway Load Balancer [Info](#)



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

Create

Create Application Load Balancer

4. Enter a **Name** for the load balancer, e.g., MyAppLoadBalancer.
5. Set **Scheme** to **internet-facing**.
6. Choose **IPv4** as IP address type.

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

balancer1

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the **IPv4** and **Dualstack** IP address types.

Load balancer IP address type [Info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address type.

☒ IPv4

Includes only IPv4 addresses.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

☐ Dualstack without public IPv4

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **internet-facing** load balancers only.

7. Choose the **VPC** and select at least two Availability Zones (AZ) subnets.

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be host premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-01f007e5fdc3847aa
172.31.0.0/16

Availability Zones and subnets [Info](#)

Select at least two Availability Zones and a subnet for each zone

☒ ap-south-1a (aps1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP addr

subnet-09a95e8b14201065f
IPv4 subnet CIDR: 172.31.32.0/20

☒ ap-south-1b (aps1-az3)

Subnet

Only CIDR blocks corresponding to the load balancer IP addr

subnet-04eb704976977f7fc
IPv4 subnet CIDR: 172.31.0.0/20

☒ ap-south-1c (aps1-az2)

Subnet

Only CIDR blocks corresponding to the load balancer IP addr

subnet-01bdf90b1522f801b
IPv4 subnet CIDR: 172.31.16.0/20

8. Select the Security Group (if created) or create a new security group.

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

SecurityGroup1
sg-0b56e355a47d036a4 VPC: vpc-01f007e5fdc3847aa

9. Select the Listeners and Routing.

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80

Remove

Protocol

HTTP

Port

80

1-65535

Default action [Info](#)

Forward to

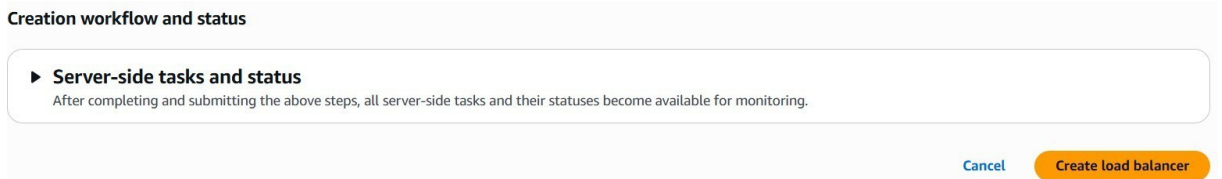
talb1

Target type: Instance, IPv4

HTTP

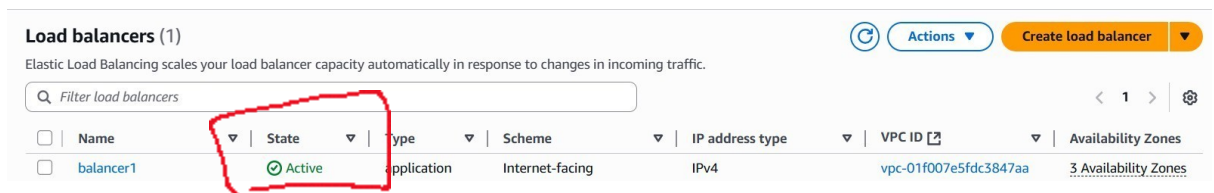
[Create target group](#)

10. Have a quick **Review** and click on **Create load balancer**.

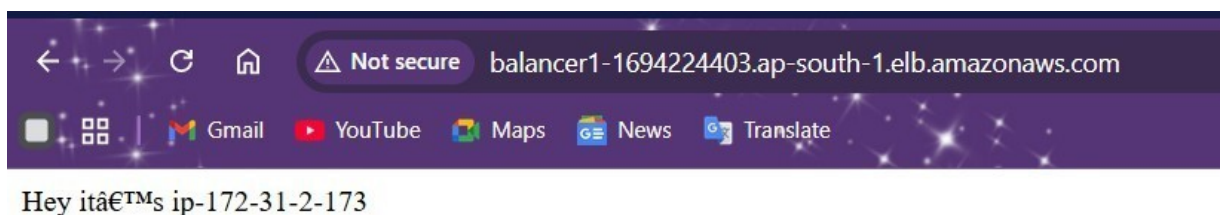


Step 6: Validate and Test

1. In the **Load Balancer List**, select the load balancer you just created.
2. Confirm the status changes from *provisioning* to *active*. When the Status is Active then only run the **DNS**.



3. Check the **Target Groups** tab and ensure the EC2 instances show as **healthy**.
4. Copy the **DNS name** of the load balancer (found in the Description tab).
5. Paste the **DNS name URL** in a browser to verify the site loads and traffic is balanced across instances. It might take few refreshes to make the DNS run correctly.
- 6.



Step 7: Clean Up (Optional)

- To avoid AWS charges, delete the load balancer, target group, security group, and terminate EC2 instances if no longer needed.